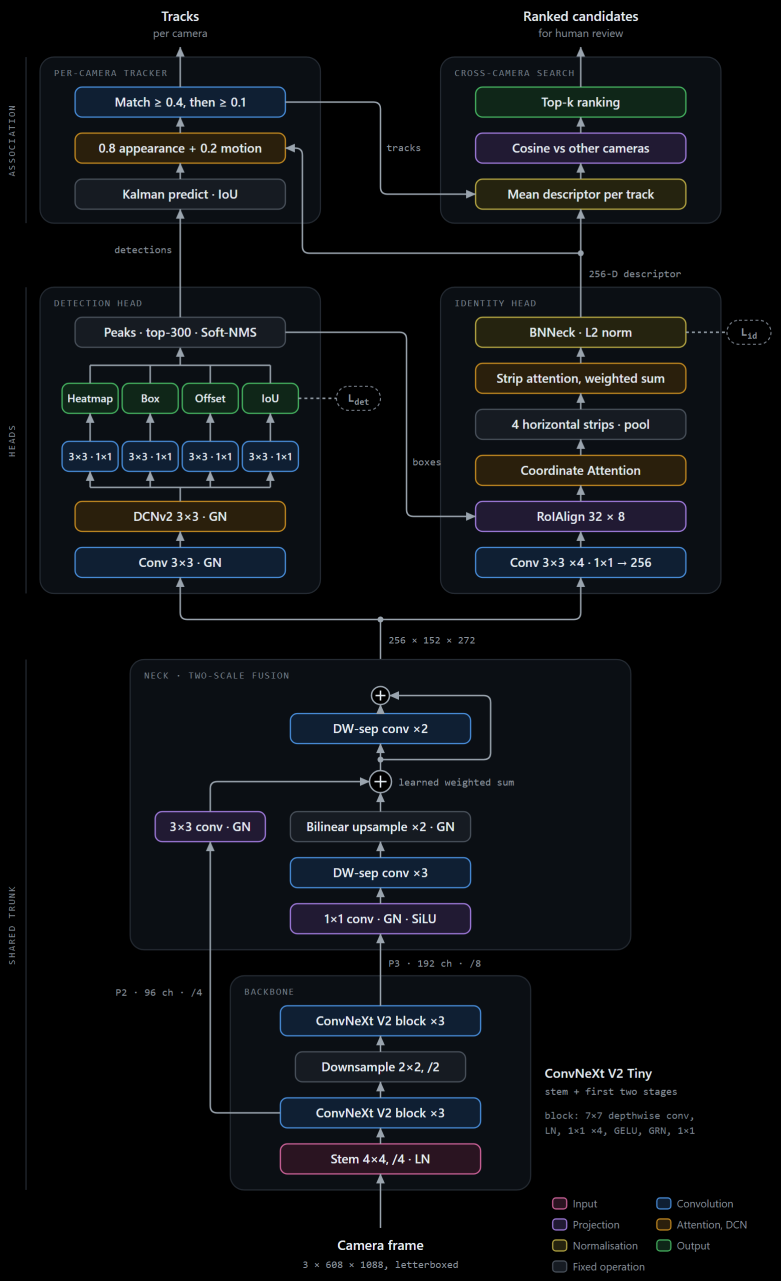
EFFIPED / MULTI-CAMERA PEDESTRIAN DETECTION, TRACKING AND RE-IDENTIFICATION

Multi-Camera Pedestrian Detection, Tracking & Re-Identification using Joint ConvNeXt V2 Architecture

One compact network finds people in each camera and describes how they look; a tracker follows them within each camera, and search ranks people in the other cameras for a person to review.

62.8%	64.08	7.78M	256-D
Cross-camera Rank-1	**MOT17 MOTA**	**Model parameters**	**Identity descriptor**
P-DESTRE validation split	MOT17 validation half	Approx. 18 FPS full pipeline	RoIAlign and four body strips

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Software: Apache-2.0
P-DESTRE-derived demonstration media: CC BY-NC-SA 4.0



01 / ARCHITECTURE

One network for detection and identity.

A frame letterboxed to 1088 x 608 passes once through the stem and first two stages of ConvNeXt V2 Tiny. A two-scale neck fuses stride-4 and stride-8 features into one 256-channel map at stride 4, read by a detection head and an identity head. The figure reads from the bottom up.

Part	What it does	Output
Backbone	ConvNeXt V2 Tiny stem and stages 1-2. Block: 7x7 depthwise conv, LayerNorm, 1x1 expand x4, GELU, GRN, 1x1.	P2 96 x 152 x 272, P3 192 x 76 x 136
Neck	P3: 1x1 conv, GroupNorm, SiLU, 3 depthwise-separable convs, bilinear 2x upsample. P2: 3x3 conv. Learned weighted sum, residual refine.	256 x 152 x 272
Detection head	3x3 conv, then 3x3 DCNv2; per output a 3x3 and a 1x1 conv: centre heatmap, box distances, offset, IoU.	4 maps at stride 4
Identity head	Own conv branch to a 256-channel map; RoIAlign 32 x 8, Coordinate Attention, 4 strips with softmax attention, BNNeck, L2.	256-D descriptor

7.78M parameters in total. GroupNorm is used throughout the neck and heads, so training works at two images per batch.

02 / TRAINING

Detection and identity balanced by learned task uncertainty.

```
L = 1/2 [ exp(-s_det) L_det + s_det + exp(-s_id) L_id + s_id ]
```

s_det and s_id are learned with the network, starting at -1.85 and -1.05. Box targets cover the centre Gaussian; identity targets sit on the centre pixel.

Detection, L_det

Term	Loss	Weight
Heatmap	Gaussian focal loss (alpha 2, beta 4); low-visibility positives weighted up to 1.5x	1.0
Box size	GIoU on left, top, right, bottom distances at every pixel inside the centre Gaussian	0.5
Centre offset	L1 on the sub-pixel offset	1.0
Repulsion	Smooth L1 on overlap with a neighbour beyond their true overlap	0.4
Box quality	Binary cross-entropy between predicted and actual IoU	1.0

Identity, L_id

Term	Loss	Weight
Identity	ArcFace (s 16, m 0.25) after BNNeck; label smoothing 0.1, dropout 0.1, visibility-weighted	1.0
Strip identity	ArcFace on each of the 4 body strips	0.3
Metric	Batch-hard triplet (margin 0.5) before BNNeck, mined from a 8,192-entry cross-batch memory	0.3
Strip metric	Batch-hard triplet on each strip, within the batch	0.09
Strip diversity	Penalises similarity between one person's strip descriptors	0.1

Setting	Value
Data	P-DESTRE, MOT17, MOT20 and SOMPT22 sequences; CrowdHuman mixed in at 12%
Sampling	Identity-aware batches, +50% cross-camera batches, 1 frame in 3, boxes at least 30% visible
Augmentation	Colour jitter, random erasing, mosaic and mixup, blur, noise, grayscale, JPEG and resolution degradation, gamma
Optimiser	AdamW, learning rate 0.0001, weight decay 0.01, backbone at 0.1x
Schedule	15 epochs, linear warm-up over 1,000 batches, cosine warm restarts (T0 5, x2)
Batch	2 images x 8 accumulation steps, gradient clip 10

Peaks become boxes, boxes become descriptors, descriptors become candidates.

Step	What happens
Decode	3x3 max-pool peaks, top 300, score = sqrt(heatmap x predicted IoU), Gaussian Soft-NMS (sigma 0.35)
Track	Kalman prediction; detections scoring at least 0.4 matched first on 0.8 appearance + 0.2 motion, then down to 0.1 on box overlap; lost tracks kept 30 frames. Appearance: 0.6 fused + 0.4 strip cosine, strips weighted by mutual visibility
Search	Mean descriptor per person; people in the other clips ranked by cosine similarity for human review

Evaluation	Retrieval	Detection	Tracking
P-DESTRE validation split, cross-camera retrieval	62.8% Rank-1	90.74% mAP@0.5	-
P-DESTRE test split, cross-camera retrieval	61.3% Rank-1	88.4% mAP@0.5	-
MOT17 validation half; min_visibility=0.3; detection threshold 0.30; NMS 0.60	-	-	64.08 MOTA / 74.24 IDF1 / 61.34 HOTA

EffiPed: 7.78M parameters, 1088 × 608 input, 256-D descriptor, approximately 18 FPS for the full pipeline on an NVIDIA RTX 5080 Laptop GPU. Single archived measurements: no repeated runs, intervals or significance tests. Performance depends on occlusion, box quality, pose, lighting, camera layout, time gaps and domain shift.

04 / IDENTITY REVIEW CONSOLE

The hosted console replays saved results; a person makes every decision.

Archived session diagnostic	Value
Synchronised cameras	4
Frames per clip	150
Camera-local tracks	367
Cross-camera groups	45
Pairwise association precision	0.77

Archived diagnostic from an earlier run over four synchronized P-DESTRE clips; separate from the current browser fixture produced with a research version of EffiPed.

Hosted demo

Precomputed, no model in the browser

Four clips, every saved detection and track, and the ranked candidates. Thresholds filter those records; they cannot create a detection the model never made.

Local service

FastAPI and CUDA inference

Uploads, job progress, search by example, asset retrieval and explicit cleanup, when an authorised checkpoint is available locally.

Responsible use

Candidates, not identifications

Similarity ranks candidates for review and is not proof of identity. Public weights are withheld until every training-data source permits redistribution.

References: P-DESTRE - arxiv.org/abs/2004.02782 | Media licence - creativecommons.org/licenses/by-nc-sa/4.0/ | Source - github.com/aswanth-07/effiped-multi-camera-tracking